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Bihar Engineering University, Patna

B.Tech 6th Semester Exam., 2026

Code: 105602 | ***Time: 03 Hours***

Semester: VI | ***Subject: Machine Learning*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven questions only): **[2 x 7 = 14]**

(a) A model has 98% training accuracy and 72% test accuracy. Which of the following is the most likely explanation?

(i) High bias
(ii) Overfitting
(iii) Underfitting
(iv) Insufficient features

(b) Which statement about hypothesis space is correct?

(i) A larger hypothesis space always gives better generalization
(ii) Hypothesis space is unrelated to inductive bias
(iii) A smaller hypothesis space always gives lower training error
(iv) A larger hypothesis space can increase the risk of overfitting

(c) Ridge regression primarily uses which penalty?

(i) L0
(ii) L2
(iii) L1
(iv) Entropy

(d) During backpropagation, the main purpose of propagating the error backward is to

(i) Generate new training data
(ii) Normalize the input
(iii) Remove hidden layers
(iv) Compute gradients for updating network weights

(e) LDA generally assumes that different classes have

(i) Different covariance matrices
(ii) No covariance
(iii) The same covariance matrix
(iv) Identical means

(f) If the p-value is smaller than the significance level α, we generally

(i) Reject H_0
(ii) Accept H_0
(iii) Increase α automatically
(iv) Reject H_1

(g) AdaBoost primarily focuses subsequent learners on

(i) Randomly selected features only
(ii) Previously misclassified or difficult observations
(iii) Only correctly classified observations
(iv) Unlabelled observations

(h) K-medoids differs from K-means because K-medoids uses

(i) An actual data point as the cluster representative
(ii) The mean of all clusters
(iii) A randomly generated point
(iv) Eigenvectors

(i) A dataset contains two elongated, non-spherical clusters. Which method is generally more appropriate than standard K-means?

(i) K-means
(ii) Linear regression
(iii) Density-based clustering
(iv) Naive Bayes

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(j) In the E-step of EM, we primarily

(i) Update model parameters

(ii) Remove missing data

(iii) Estimate the latent-variable responsibilities

(iv) Calculate classification accuracy

Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Explain the bias–variance trade-off and discuss how model complexity affects underfitting and overfitting.

(b) Explain the basic concepts of Reinforcement Learning. Also explain the exploration–exploitation trade-off and the role of the ε-greedy strategy.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) A test is applied to 1000 patients: 200 are sick, and 800 are healthy. It correctly detects 160 sick patients, while 40 healthy patients test positive. Construct the confusion matrix and calculate accuracy, precision, recall, and F1-score.

(b) The following data shows the hours studied (X) and marks obtained (Y) by 6 students:

X	2	3	5	6	8	9
Y	50	55	65	70	80	85

Find the regression equation of Y on X and predict the marks for a student who studies 7 hours.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Explain Principal Component Analysis. Describe the steps involved in transforming the original features into principal components.

(b) Explain Bayesian Learning in detail. Discuss Maximum Likelihood, Maximum A Posteriori and Bayes estimation, and explain the role of conjugate priors with suitable examples.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Explain the Perceptron learning algorithm. Describe the weight and bias update rules and illustrate the algorithm with a suitable example.

(b) Explain Support Vector Machines. Discuss the concept of maximum-margin hyperplane, support vectors, and soft margin.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Using the ID3 algorithm, construct a decision tree for the given dataset.

Outlook	**Temperature**	**Humidity**	**Wind**	**Play**
Sunny	Hot	High	Weak	No
Sunny	Mild	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes

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(b) Given the following 2D points: A(1,1), B(1,2), C(2,1), D(2,2), E(8,8), F(8,9), G(9,8), H(15,15). Using DBSCAN with ε = 1.5 and MinPts = 3, identify the core, border and noise points and determine the clusters.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Explain ensemble learning and its motivation. Why can combining multiple weak or unstable learners improve prediction performance?

(b) Explain the Random Forest algorithm. Discuss the role of bootstrap samples and random feature selection.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Explain the complete K-means algorithm. Discuss its objective function, convergence criterion, and major limitations.

(b) Explain hierarchical clustering. Differentiate between agglomerative and divisive approaches and explain how a dendrogram is constructed.

Q.9 Write a short note on any two of the following: **[7 x 2 = 14]**

(a) Q-Learning

(b) Bayesian Networks

(c) Naive Bayes classifier

(d) Bagging and Boosting

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